You will undertake a personal research paper as a major component of our class for the final project. The topic must be chosen in consultation with me (via a proposal) and should be focused on a relatively new statistical topic to you (perhaps you heard the name in a class, but didn't study the method, or didn't study the method in depth). This project provides an opportunity for you to demonstrate achieving some of the learning goals for the statistics major. As such, successful completion of the project is required to pass the course (which, in turn, serves as the comprehensive evaluation for the statistics major). The final deliverable for the project includes a report written in RMarkdown (via .Qmd) (alternatives possible if you are using Python). You must have a polished report with visible code and proper formatting. You may include other deliverables (Shiny apps, supplemental files, etc.) as appropriate.

Assignment

In the learning goals for the major, you can see that we expect graduates of the program to be able to understand statistical techniques in primary literature and to be able to communicate their understanding about statistics, as well as have the ability to identify resources to expand and fill in gaps in their knowledge. In this project, you will:

1. Identify a new statistical topic of interest to you. (Process started with class activities.)

2. Research the topic, identifying multiple sources to assist you in learning about it.

3. Prepare an expository review of the topic at a level that a classmate could understand. (Details below. Imagine you are charged with teaching a peer the topic.)

4. Demonstrate your understanding of the topic with application(s) to a data set and/or simulations.

5. Communicate your findings/results of the application(s)/simulations.

6. Collate the exposition and findings/results of the application(s)/simulations into a final report.

This project is designed to challenge you. Learning about a simple extension of a technique and applying it to a data set that you downloaded in .csv format is not acceptable. You should seek to challenge yourself and demonstrate your statistical analysis skills. The project is deliberately open-ended to allow you to fully explore your creativity - you can have more deliverables than the final report - think about Shiny apps or example simulation code that could be shared with a wider audience. Consider using data sets that require considerable wrangling skills to let you tackle a problem of interest to you with a new technique. You can easily combine the exposition of the new technique with a comparison to another technique (learned in our class or another course).

Note that "toy" examples (including ones you generate, such as the one used in our p-value adjustment example) may be used to demonstrate techniques, but do not count as an application or simulation. Toy examples may be included (this is encouraged!) for the exposition but the idea for the application is to try out the technique on more complicated data, and for simulations, to do more than generate a single data set (or do a single run of something, etc.). You'd want a simulation study where you were assessing something if going that route.

There are five main rules that must be followed:

1. The project is individual.
2. The statistical topic must be new to you. That means something you did not formally study in-depth in a class, although you can study variants of such methods, or propose going into detail about a method that you studied briefly. If you learned about a topic during an internship or other experience, please consult with me to see if it is appropriate. If you have already applied the technique to a data set previously, it is likely not appropriate. What is new for one student may not be new to another - you've taken different courses and had difference experiences. I must agree that the topic is new to you, which is why your proposal has you describe what you already know about it.
3. You must prepare a paper/report as part of your final submission.
4. The paper/report must include 1) an exposition on the new topic showing that you were able to research the topic appropriately and can explain it at a level of detail such that a peer could understand and 2) either application(s) to a data set beyond a simple "toy" example, or simulations to demonstrate your understanding of the technique.
5. The entire report/analysis, etc. must be reproducible, in line with our learning goals for the statistics major.

1 Components and Organization

In order to help you tackle this project, there are a number of components / deliverables designed to be turned in along the way. These are described roughly below. Details will be provided with each particular assignment / component.

1.1 Github

In your personal repo for the course, you should create a "FinalProject" folder (or something similar) in which to keep all relevant project materials. Remember that Github has a MB upload limit. Don't try to upload anything over 50 MB (100 MB will fail). When you get to the data for your application (if going that route), consult with me for alternatives to an upload for a large data set. You should put this assignment file and all future work on the project in the appropriate folder. You will also be asked to upload some materials to Gradescope for review. A final copy of your paper will complete your portfolio in the repo (you will end up with 2 copies - one in the FinalProject folder and one in your Portfolio folder). Remember that maintaining your repo and keeping things organized, as well as doing all portfolio assignments for the semester, is a separate part of your course grade.

1.2 Brainstorm Topics - Informal Presentation

We are undertaking the brainstorm as a group exercise, with informal presentations in class on Monday, Nov. 3rd. See separate instructions. Everyone is responsible for preparing two slides, one of each of their assigned topics, to share with the class. Preparing these slides should take you a max of 30 minutes each (quite possibly much less time if you saw the methods in a class).

1.3 Proposal

After the topic brainstorm/discussion, you will select and refine a topic and prepare a proposal for your topic/report. A proposal template will be provided that you can fill in and submit in Gradescope. Revisions may be needed for clarification, and will be handled individually. There is no formal revision re-submission date, since these vary too much individually. However, your final accepted proposal (including all revisions) should be kept in your Project folder. It's laid out in a way to assist you with the project - so you want it there for your reference (and mine, when assisting you).

The remaining components are designed to help you with researching the topic, writing the paper/report, and staying on track to complete the final product. Bear in mind that while researching the topic for the exposition, you can (and should) start looking for your data, getting it organized/wrangled, etc. if doing an application, or start outlining your simulation plans, if going that route. The deliverables are not focused on those components - as you have considerable experience with those already. But you need to factor this into your project planning (which we'll address in a project planning activity).

The proposal deadline is Wednesday, Nov. 5th by midnight.

1.4 Annotated Bibliography

You will need to assemble multiple sources to learn about your new topic. (The assignment sheet for this will have more details about how many and what type are recommended, etc.) You will then construct an annotated bibliography to help in construction of the exposition portion of the report. The annotated bibliography is submitted as a component in Gradescope, and is due Thursday, Nov. 13th by midnight. This will help you learn about your technique from multiple sources and begin to organize your expository review of the topic.

1.5 Exposition

The portion of the paper where you describe the new technique as though you were teaching it to a classmate (with whatever name you want to give it - Literature, Background, or Exposition) will be based on your annotated bibliography. You can add sources to what you already have in the annotated bibliography (especially if my feedback says you need to!). To be sure that you are on track with the exposition, a full draft of this portion of the paper is due the Friday before Thanksgiving break. This will also help with incorporating proper citation early in the writing process, as you'll have a bibliography that we can implement for this portion of the paper at a minimum.

What should you think about for the exposition? Imagine you are teaching a classmate about your topic. Envision a coherent and comprehensive lesson plan. For example,

1. What does your reader need to know or be reminded of before you can introduce the topic?

2. What is the topic and what is it used for?

3. What are the technical details that a reader would need to understand if planning to pursue this themselves?

4. How does it work? Can you make connections to other topics the reader will know more about?

5. Where has the topic come up before? (Giving examples of usage is extremely helpful to someone learning a new topic.)

Your example application or simulation study would then follow this, rounding out the "lesson", though that part doesn't need to be turned in at this point. You can, however, submit as much as you want, and I'll give feedback on at least the exposition to everyone (and more if I can get to it, for everyone who turns in more).

1.6 Discussion on Exposition

So that you can practice communicating about your topic (and to let me ask any clarifying questions and help give feedback before the final submission), you'll have an oral discussion with me the week after Thanksgiving break to discuss your exposition / understanding of your statistical topic. This will be similar to our other discussions (mostly like the data analysis project one) with it's own rubric. Feedback here and from what I return from your drafts can be used to improve your final submission.

1.7 Complete draft for peer review and peer review form

A complete draft of your paper/report for peer review is due by midnight on Friday, December 5th to your repos. You will be placed in a group of 4 (assuming numbers work out) for peer review. You will only review ONE other submission (pairs within a quad) but you will have access to all submissions in your group so you can see other submissions for ideas on how your classmates have structured things, etc.

I will organize submissions and groups on Saturday the 6th via Google Drive. You will have a form template (a Google doc to fill in, not officially a "form") to complete for the submission you are reviewing. Completed forms should be in the Drive so they are shared with your groupmates before class on Monday, December 8th. That class day will be devoted to sharing feedback with each other. This is an important contribution to helping your classmates with their reports. For example, if you don't understand something in their exposition or in their application/simulation section, you have an opportunity to help them improve their paper by telling them so, and offering suggestions for improvement!

The writing center folks helped me develop this activity in the past. They recommend setting 2-4 hours aside to complete the review for your peer. So, it's important to set this time aside on your weekend, and also, reinforces that I need the drafts on time so you all get them by like Saturday at noon, at the latest (I'll try to have it set and emailed sooner).

1.8 Check-in

A check-in is due by the end of the day on the last day of classes, Wednesday, Dec. 10th, to check in on what work you have left to complete before the paper is finished. You will need to create an issue in your course Github repo with answers to the following:

1. What is the single biggest unresolved issue you are having? Please describe it briefly, and what your plan is for resolving this issue.
2. What else is left to be completed before the paper/project is finished?
3. What is your proposed timeline to complete those items?
4. Do you have any questions for me? If so, what are they?

I will review these and reply, closing the issue if everything looks fine, or leaving it open if you need to see a reply from me about a question you asked (you can then close it once you've read it) or if we need to continue discussion.

1.9 Report

Your final report and any other deliverables will be due by midnight on the first day of exam period, Monday, December 15th. This will include a separate integrity page that you put in the repo to list sources you used to adjust things in the report, specify any generative AI use, etc. The sources used for the statistical material and data set (if applicable) must be cited in your report.

All materials should be in your repo folder for the project. A final copy of the report should be uploaded to the corresponding Gradescope assignment and a copy of the pdf put into your Portfolio folder.

I expect to prepare a template for the report, etc. as the semester proceeds. However, you may each need to adapt it based on your topic, plans for application(s)/simulations, and deliverables.

IMPORTANT: There are NO extensions for this project. We are devoting almost an entire month of class time to it, and your repo should show your progress along the way. The final submission for the report (pdf) must be in Gradescope by midnight. It must show all your code for a fully reproducible analysis. You have some flexibility for getting any lingering files in the repo because I will not pull those until Tuesday morning. But I can see what comes in after midnight due to commit times and it should ONLY be things like a lingering file you didn't realize was missing, changing a file path because you didn't realize something wasn't reproducible and you are fixing it, or creating a readme to explain what files are what. In other words, MINOR changes. Why does this matter? Submitting something late (or trying to) is, in effect, giving yourself an extension without permission that your peers didn't have. So, we keep the same deadline for everyone. If it's close to the deadline and you aren't quite finished polishing something, submit what you have BY the deadline. Make a commit (and push so you don't forget) before the deadline. Then, if you keep working, I will be able to see what was done on-time and what was after. The Gradescope report version is what gets graded for the report component, so it must be on-time.

1.10 Portfolio Reflection Discussion

On Wednesday, December 17th and Thursday, December 18th, we'll hold a final oral discussion to round out the semester. This one will be like the first one this semester, primarily reflection on your part, with some ability for me to ask any clarifying questions on your submission (which I'm planning to review Tuesday). You won't need to "study" for it (just bring your laptop to access your submission and portfolio) and it will take the place of a written reflection. Details, etc. given like for the other discussions closer to time.

2 Assessment Items and Other Information

In this section, the goal is to outline all the various assessment items for the project. Minor adjustments may occur as I work to integrate the two discussions into this framework, but won't introduce any surprises (I have to figure out points/weights of things).

2.1 Deliverables Other than the Report

For submitted deliverables other than the report, you should expect that they have an "on-time" component (usually 5 points) and a "completeness" component (more points). For example, is your proposal submitted on-time? Is it completely filled out / all necessary information provided and is it filled out well? In other words, you provided sufficient detail for me to understand and assess your proposal. (Same thing for your slides for the topic brainstorm, annotated bibliography, check-in via the repo, separate integrity page, etc.) This would further apply to any additional deliverables you decide to add, like if you make a shiny app, it should be "on-time" and "completed" well as described in your proposal.

For the exposition draft, it will have an additional component related to the statistical foundation/material. Is the statistical material presented correctly? Any issues, etc. Note that these can be addressed before the final submission (which will also have a similar category, and the idea is you should be able to improve if you address comments given to you).

For the peer review, the draft must be "on-time" and you must complete the review "on-time" and completely for your peer. Your draft submitted is not assessed on completeness or statistical foundation by me. But you should submit as much as you can so you can get the best feedback possible from your peer.

The discussions will have separate points (not on-time and completeness). The exposition discussion will have a rubric somewhat similar to the data analysis rubric, but focused on presentation of the method (not method and results) and explanation. Likely only two categories as well. So presentation/explanation during discussion and communication clarity/confidence. The final reflection discussion will have a rubric similar to that from the first discussion on the tailored project. Have not made up my mind about number of categories, etc. there yet.

2.2 Report Assessment Criteria

These are the major categories to expect on the rubric for the report. For each of these, imagine they are being split into the Exemplary, Proficient, Developing, and Beginning categories with points allotted within each. Some may need split into two sub-categories, etc. Still working on that. This is the same framework as used in past years in terms of the categories below.

- Understanding of Topic: Does the paper demonstrate strong understanding of the topic? Are there any issues with the statistical content in the development of the topic? Does the write-up show appropriate synthesis of material from multiple sources? Do descriptions of example applications from literature demonstrate appropriate technique use? Does the exposition provide a comprehensive overview of what the reader needs to know about the topic?

- Communication to Audience: Does the paper introduce the topic well to the audience? Does the writing help communicate the author's statistical knowledge to a peer? Are there appropriate definitions, background, development, and examples to communicate understanding of the topic? Does the audience come away with an appropriate sense of the topic and its use/application, including theoretical concerns, such as conditions (as appropriate)?

- Application(s) / Simulations - Statistical Content: Are the application(s) / simulations performed appropriately from a statistical standpoint? Are there any issues with the analysis or simulation study setup? Are applications and the simulation study described well enough to be reproducible? Are statistical issues encountered addressed?

- Application(s) / Simulations - Write-up: How well does the write-up convey the findings/results of the application(s)/simulations? Are data sets and simulations described in enough depth for a peer to follow along? Is appropriate support (figures / output, etc.) provided? Are tables / graphics / figures crafted well (captions, look nice, etc.)? Is each table / graphic / figure explained and interpreted well enough that a peer can understand the data it is presenting, as well as its purpose/relevance to the point it's meant to illustrate?

- Presentation / Organization: Is the paper organized well? Are ideas presented in a logical order with a guiding voice / narrative thread that makes clear to the reader what each section is doing, the relationship between the sections, and the overall logic of the paper? Has the paper been proof-read and run through spell-check?

- Code: Is the code reproducible? Is all code shown, readable, and formatted well? Does all code (and associated comments) fit on the pages? All code should be visible, except for the initial setup chunk where packages are loaded.

- Citation / Use of Sources: Are sources incorporated appropriately into the text? Are attributions for direct quotes and figures included? Are sources listed appropriately in a bibliography at the end of the report? Data sets (for applications) must be cited. R packages (especially if introducing a new one) have citations too! Citations must be in-line and at the end of the report. Footnotes are not appropriate (the text is too small for this to be inclusive). The template demos appropriate citation for you.

Remember there is a separate integrity page where you will put sources used to assist in constructing the report, such as for fixing code errors, learning how to do something in LaTeX, etc. The citation listed here is for citation within the report - the materials from your annotated bibliography and for data sets you are using, etc.

2.3 Commit Behavior

As with our previous projects, reasonable and responsible commit behavior will be considered an assessment item. What does this look like for this project?

At an absolute minimum, every submitted deliverable (including the ones submitted to Gradescope) should have their own "final" commit, and say, half of our class days (with more than half-time) working on the project should have their own commits as well (I will remind you in class). Commit messages may be basic. This is beginning level (as the rating). If this level is not achieved, that is unacceptable commit behavior.

Beyond that, making sure you have commits for every day that we work on the project in class, raises it to developing level. Commit messages may be basic.

Moving to proficient level means making sure that your work sessions are documented well, basically a commit every time you do something meaningful for the project. You could imagine this is committing (and pushing) at the end of every working session on the project, at a minimum. It's possible to do multiple meaningful things in a single working session, resulting in multiple commits. Commit messages may be basic or slightly improved. You will need to work on the project outside of class, so this is making sure all work sessions are reasonably documented. (Remember in STAT 231, when you were introduced to Github, you were reminded to commit after every problem for homework. This is that level.)

Finally, exemplary (level) means proficient with good/stronger/better commit messages, so someone could track what you were doing by looking back through the commit history pretty well. I will admit my own commit behavior does not always rise to exemplary, so I will work on this with you!

2.4 Submitting the Project Elsewhere

Some students may be interested in submitting their final projects to USPROC, a competition for undergraduate statistics projects. More information can be found here: https://www.causeweb.org/usproc/. If you are interested, final projects for this class would qualify for the USRESP category of the competition. Projects may need some revision (for length) and code refinement for submission to the competition. Doing this bears no impact on your project assessment. I will learn about it because you have to list a faculty sponsor / where the project came from. The only way assessment from class might factor in is if more than four students from the class choose to submit, I will be asked to pick the four that should be judged for the competition, and I would select stronger submissions. But that wouldn't adjust the assessment received for class.

2.5 Generative AI Guidance

There is no requirement to engage with generative AI for this or any course activity.

If you choose to engage with generative AI for any part of this project, remember to follow the class framework. Generative AI may be used only as a thought partner, and only in places that don't jeopardize your learning/our course learning outcomes. Here, remember you are working to show you can enhance your own statistical understanding by learning a new technique and communicating that understanding at a level a classmate can understand.

You should immediately export, save, and commit any chats with generative AI about the project as you have them to keep a complete record to submit (Sharing a link has worked as well).

Adding the information to the integrity page as you go is also strongly recommended, rather than trying to do it at the end. You could keep a running pdf addendum of all the chats for easy reference too. Programs that merge pdfs (like Adobe Acrobat) can be useful for that.

Examples of what generative AI might be useful for in the context of the project and that are in line with the class framework are:

1. Asking AI for help coming up with a timeline that works for the project.

2. Asking AI for help outlining a particular section, where you already have some points you want to include.

3. Asking AI for help with a code error, making sure you can understand how it fixed it and could re-explain it later.

4. Asking AI to help fix an issue with a plot you are having, as long as you can explain how to fix it yourself later.

5. Asking AI to tighten up writing in a particular paragraph or suggest improvements in a section for clarity. (Remember to review it to be sure the AI didn't add anything it shouldn't have!)

Note how these are uses that you have available human support for, without concerns for academic integrity. Me, your peers, and the SLC (Strategic Learning Center) are resources for learning how to make timelines. Google (or any other search engine), me, and your peers are good resources for code issues or formatting challenges. Outlining and writing skills have support in the Writing Center, or you could ask a peer to review a submission.

Contrast that with the uses below, where academic integrity concerns arise. These show issues where the AI use is jeopardizing class learning outcomes / your learning experience. Obviously, the uses below (and others like them) are not allowed.

Examples of how generative AI might be used in the context of the project but that are not in line with the class framework (and therefore, not allowed):

1. Asking AI to write the exposition for you.

2. Asking AI to do all the data wrangling or write all your simulation code for the project.

3. Asking AI to draft the application or simulation section for you.

4. Asking AI to do the peer review for you.

Neither of these lists are exhaustive. Please refer back to the course syllabus or talk to me if you want to engage with Generative AI tools and have questions about appropriate use.

3 Supporting Class Activities

This is a major research project, and is paired with the learning outcome demonstrating you can enhance your own statistical knowledge. You may not have experience undertaking such a project in any subject, and likely not in statistics unless you took a 400-level elective, independent study course, or are writing a thesis. So, what will we be doing in class to support success in the project? Let's look at the series of activities planned. On any day where there is an activity, if it doesn't take the whole class period (many won't), you will be able to work on the project for the remainder of the time.

3.1 Project Introduction and Topic Brainstorm

The project is introduced fully in class on Friday, Oct. 31st with the topic brainstorm starting earlier that week, and with informal presentations in class on Monday, Nov. 3rd to help you pick topics.

3.2 What is a Simulation Study

This will be the second part of class Friday. We will look at an example simulation study (and define what that is) and how you might go about one if you are going that route. While not applicable to everyone, as many folks will pick the application route for the project, simulating is a valuable skill for a statistician, so the exposure here will be good.

3.3 Project Planning Activity

The proposal is due Wednesday, Nov. 5th by midnight, so that morning, in class, we will do a project planning activity. This will assist you with part of the proposal (your project timeline) and serve as time you can ask me questions as you prepare your proposal.

3.4 Annotated Bibliography

On Friday, Nov. 7th, we will discuss what the annotated bibliography deliverable is, what you need to do for it in depth, and appropriate strategies. This is basically a discussion of how to find sources, learn about your new topic from them, and keep track of what was useful for what. This deliverable is due about a week later. After discussing the assignment, you can work on it for the rest of the period.

3.5 Reading Literature

This is expected to be a short activity on Monday, Nov. 10th to assist you in strategies for reading the literature that you are finding for your annotated bibliography.

3.6 Presentation Topic Share

This is expected to be a short activity on Wednesday, Nov. 12th where we simply share our topics with each other as a class. You may find someone else is working on your topic (or a related one) and want to touch base to compare sources, etc. later.

3.7 Paper Writing and SMART Goals

This may be a slightly longer activity on Friday, Nov. 14th to discuss strategies for writing papers and will also introduce SMART goals, since we'll be using those during the next two weeks to help set goals for our class working periods.

3.8 Class Working Time

The full week of class before Thanksgiving break and two class days after it are designed to be spent on the project with no major activity planned (will use SMART goals form). The form allows you to ask me questions after every class session, and we have some fun with it too.

Important: If you know you will miss a class during this time due to an interview, or are sick and miss one, etc. you can make it up by having a 50 minute work session on the project on your own and filling out the form based on that session. I pull the form shortly after every class to check for new responses with questions so that I can reply to them, so I may not see questions you asked until the next pull, but can see that you've made up the time and still respond to your questions this way.

You can use the form outside of class as well, if you find it useful, as much as you want. Again, I would not see questions until the next pull, unless you want to email me to tell me to go look. If you don't like the form, you'll need it use it on the designated class days but not otherwise. In the past, some students liked it and used it a lot outside of class, while others really didn't like it, so I'm aware not everyone is a fan.

3.9 Oral Discussion on Exposition Topic

The oral discussion will take place the week after Thanksgiving break to give you feedback on that portion of the submission.

3.10 Peer Review Setup

On Friday, December 5th, we'll spend part of the class time getting set up for the peer review (a lot of logistics above), and the rest on projects.

3.11 Peer Review Discussion

Monday, Dec. 8th's class will be devoted to the peer review discussion. In the past, there have been a few minutes for folks to consider the feedback on their own after the discussion, but not a lot of project working time is expected.

3.12 Final Logistics

Our last class day will begin with final logistics, and wrap-up with course evaluations. I will try to check in with everyone in between those two items (or maybe start with the check-in), but your formal check-ins are due that evening. Not a lot of project work time is expected.

4 Timeline with Deliverables

All deliverables should be in your repo but some also have Gradescope submissions.

This timeline is trying to incorporate class activities with deliverables. Feel free to use it to aid in creating a timeline for yourself for other project aspects.

Activity or Deliverable	**Timeline**
Final Project Introduced Fully	in Class on Friday, Oct. 31st
What is a Simulation Study Activity	in Class on Friday, Oct. 31st
Slides and Informal Discussion	by Class (and in Class) on Monday, Nov. 3rd
Project Planning Activity	in Class on Wednesday, Nov. 5th
Project Proposal Due	by midnight on Wednesday, Nov. 5th (Gradescope)
Annotated Bibliography Activity	in Class on Friday, Nov. 7th
Revised Proposal	As needed
Reading Literature Activity	in Class on Monday, Nov. 10th
Topic Share Activity	in Class on Wednesday, Nov. 12th
Annotated Bibliography Due	by midnight on Thursday, Nov. 13th (Gradescope)
Paper Writing Activity	in Class on Friday, Nov. 14th
Project work week	Week of Nov. 17th
Exposition Draft Due	by midnight on Friday, Nov. 21st (Gradescope)
Project work week	Week of Dec. 1st
Exposition Discussions	Week of Dec. 1st
Setup for Peer Review	in Class on Friday, Dec. 5th
Draft for Peer Review	by midnight on Friday, Dec. 5th (pdf in repo)
Peer Review	on Weekend of Dec. 6th and 7th
Peer Review Discussion	in Class on Monday, Dec. 8th
Final Logistics and Evaluations	in Class on Wednesday, Dec. 10th
Check-in	by midnight on Wednesday, Dec. 10th (repo)
Final Report	by midnight on Monday, Dec. 15th (Gradescope)
All remaining deliverables	by midnight on Monday, Dec. 15th
Reflection Discussion	Wednesday or Thursday December 17th or 18th